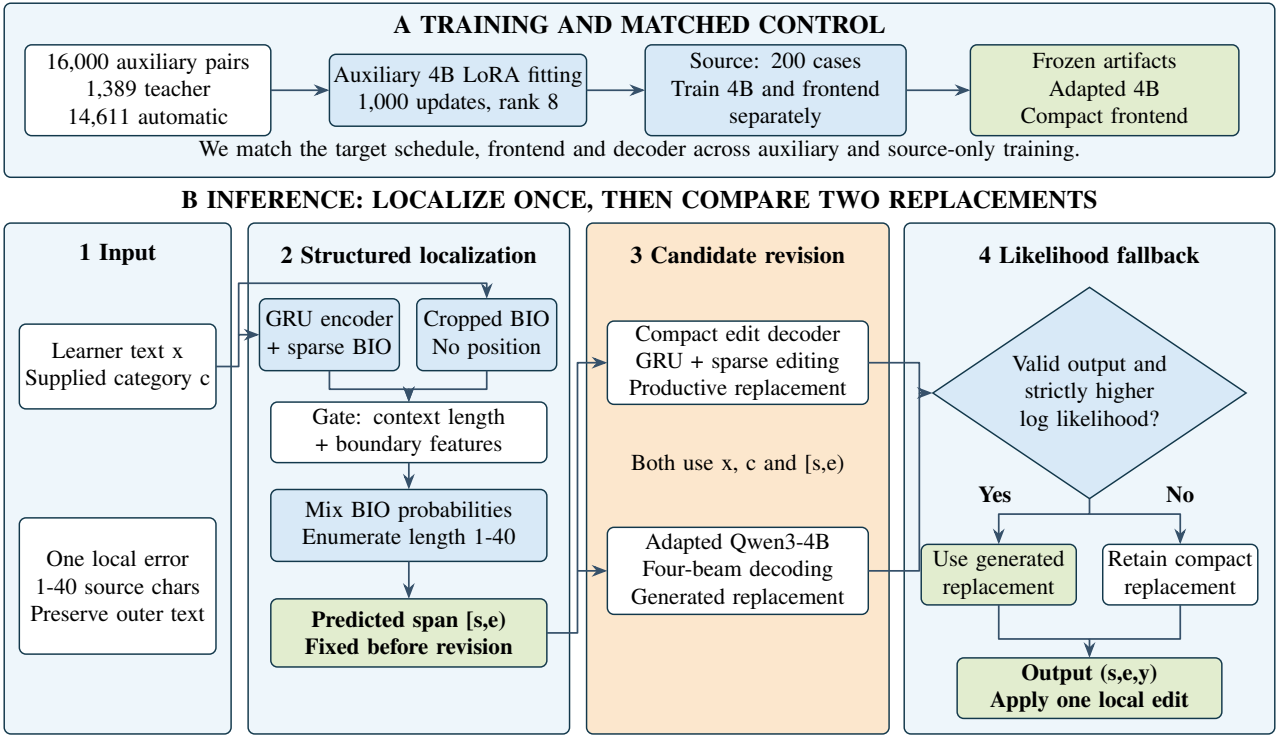
A TRAINING AND MATCHED CONTROL
16,000 auxiliary pairs
1,389 teacher
14,611 automatic
Auxiliary 4B LoRA fitting
1,000 updates, rank 8
Source: 200 cases
Train 4B and frontend
separately
Frozen artifacts
Adapted 4B
Compact frontend
We match the target schedule, frontend and decoder across auxiliary and source-only training.
B INFERENCE: LOCALIZE ONCE, THEN COMPARE TWO REPLACEMENTS
1 Input
Learner text x
Supplied category c
One local error
1-40 source chars
Preserve outer text
2 Structured localization
GRU encoder
+ sparse BIO
Cropped BIO
No position
Gate: context length
+ boundary features
Mix BIO probabilities
Enumerate length 1-40
Predicted span [s,e)
Fixed before revision
3 Candidate revision
Compact edit decoder
GRU + sparse editing
Productive replacement
Both use x, c and [s,e)
Adapted Qwen3-4B
Four-beam decoding
Generated replacement
4 Likelihood fallback
Valid output and
strictly higher
log likelihood?
Yes
No
Use generated
replacement
Retain compact
replacement
Output (s,e,y)
Apply one local edit
Validity: changed, nonempty, bounded length, correct end token, no line break. Ties retain the compact output.